QUIZ 6 /COAL FALL 24

Name: ______________________ Roll No: ____________ Section: ______

1. Fill in the Direct-mapped cache for the address given below. Main memory is given on board. Calculate

QUIZ 6 /COAL FALL 24	18/11/2023	Marks: 10 / Time: 30 min
Name: ______________________	Roll No: ____________	Section: ______

1. Fill in the **Direct-mapped cache** for the address given below. Main memory is given on board. Calculate Hit and Miss Rate

M M M M
12H, 1DH, 3AH, 25H, 10H, 1FH, 38H, 27H, 13H, 1EH, 3BH, 11H, 24H, 26H, 39H, 1EH, 1CH, 30H, 27H, 12H.
H H H H H H H H H H H H H H H H

TAG		00	01	10	11	V
01	00	8B	56	56	65	~~0~~1 1
10	01	3E	9F	77	89	~~0~~1
11	10	45	65	35	98	~~0~~1
01	11	12	39	75	A6	~~0~~1

HIT RATE = 80% **MISS RATE =** 20%

2. Fill in the **Direct-mapped cache** for the address given below. Main memory is given on board. Calculate Hit and Miss Rate

M M M M M M M M M M M M M M M M M M M M
20H, 33H, 1CH, 3FH, 21H, 32H, 1DH, 3DH, 22H, 31H, 1EH, 3CH, 23H, 30H, 1FH, 3EH, 23H, 30H, 1DH, 3CH

TAG		00	01	10	11	V
[illegible]	00	B3 2A B3 2A	3C 3B 3C 3B	3F 37 3F 37	3D 9D 3D 9D	1
	01					
	10					
[illegible]	11	[illegible] 37	39 3F 39 3F	[illegible]	A6 6C A6 6C	1

HIT RATE = 0% **MISS RATE =** 100%

3. Design Cache for direct Mapping? If your Main Memory address size is 8-bits?
Note: Tag is 1 bit, Index/line is 3 bits, offset is 3 bits?

Tag	Set	000	001	010	011	100	101	110	111
	000								
	001								
	010								
	011								
	100								
	101								
	110								
	111								

QUIZ 6 /COAL FALL 24 18/11/2023 Marks: 10 / Time: 30 min

Name: ______________________ Roll No: ______________ Section: ________

4. Fill in the **2 Way Set Associative Cache** for the address given below. Main memory is given on board. Calculate Hit and Miss Rate.

M M M M M

20H, 33H, 1CH, 3FH, 21H, 32H, 1DH, 3DH, 22H, 31H, 1EH, 3CH, 23H, 30H, 1FH, 3EH, 23H, 30H, 1DH, 3CH

H H H H H H H H H H H H H H H H

TAG		00	01	10	11	V	LRU
100	0	B3	3C	3F	3D	1	[illegible]
110		2A	3B	37	9D	1	[illegible]
011	1	12	39	15	A6	1	[illegible]
111		37	3F	3E	6C	1	[illegible]

HIT RATE = $\frac{16}{20} \times 100 = 80\%$ MISS RATE = $\frac{4}{20} \times 100 = 20\%$

5. Fill in the **2 Way Set Associative Cache** for the address given below. Main memory is given on board. Calculate Hit and Miss Rate.

M M M M M M M M M M M M M M M M M M

13H, 38H, 08H, 12H, 39H, 09H, 11H, 3AH, 0AH, 10H, 3BH, 0BH, 11H, 38H, 09H, 13H, 38H, 08H

TAG		00	01	10	11	V	LRU
[illegible]	0	[illegible]	[illegible]	[illegible]	[illegible]	1	[illegible]
[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	1	[illegible]
	1						

HIT RATE = 0% MISS RATE = 100%

6. Fill in the **4 Way Set Associative Cache** for the address given below. Main memory is given on board. Calculate Hit and Miss Rate.

M M M

13H, 38H, 08H, 12H, 39H, 09H, 11H, 3AH, 0AH, 10H, 3BH, 0BH, 11H, 38H, 09H, 13H, 38H, 08H

H H H H H H H H H H H H H H H

TAG		00	01	10	11	V	LRU
010	0	8B	56	5C	65	1	[illegible]
111		45	65	35	98	1	[illegible]
001		00	12	35	45	1	[illegible]
	1						

HIT RATE = [illegible]% MISS RATE = 16.66%